



Acidithiobacillus ferrooxidans and its potential application

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Abstract

The widely distributed *Acidithiobacillus ferrooxidans* (*A. ferrooxidans*) lives in extremely acidic conditions by fixing CO₂ and nitrogen, and by obtaining energy from Fe²⁺ oxidation with either downhill or uphill electron transfer pathway and from reduced sulfur oxidation. *A. ferrooxidans* exists as different genomovars and its genome size is 2.89–4.18 Mb. The chemotactic movement of *A. ferrooxidans* is regulated by quorum sensing. *A. ferrooxidans* shows weak magnetotaxis due to formation of 15–70 nm magnetite magnetosomes with surface functional groups. The room- and low-temperature magnetic features of *A. ferrooxidans* are different from other magnetotactic bacteria. *A. ferrooxidans* has potential for removing sulfur from solids and gases, metals recycling from metal-bearing ores, electric wastes and sludge, biochemical production synthesizing, and metal workpiece machining.

Keywords *Acidithiobacillus ferrooxidans* · Genomovar · Iron and sulfur oxidation · Quorum sensing · Magnetic property · Potential application

Introduction

Acidithiobacillus ferrooxidans (*A. ferrooxidans*), originally named *Thiobacillus ferrooxidans* (*T. ferrooxidans*), was first isolated by Colmer and Hinkle in 1947 from acid mine drainage (Crundwell 1997). It is a Gram-negative, chemotrophic, acidophilic aerobe (Yan et al. 2010b). Additionally, it utilizes CO₂ as the only carbon source, depends on the oxidation of Fe²⁺ or S, and reduced sulfur compounds to obtain energy (Temple and Colmer 1951; Valdés et al. 2008). During the Fe²⁺ oxidation process, most of the electrons are transferred to O₂ along the potential gradient, while a small amount of electrons are transmitted inversely along the potential gradient (Barrie and Hallberg 2009). The

NAD(P)H generated in the latter process is involved in the fixation and aerobic metabolism of CO₂ (Barrie and Hallberg 2009; Valdés et al. 2008). The two electron transfer paths, namely the downhill potential gradient and the uphill potential gradient, are interconnected (Barrie and Hallberg 2009). Sulfur oxidation is coupled with high redox potential in the electron transport chain, which is important for fixing CO₂ by *A. ferrooxidans* (Barrie and Hallberg 2009). Compared with the Fe²⁺ oxidation system, the sulfur oxidation process is more complex and research progress into this system is relatively slow.

A. ferrooxidans regulates adaptation to the environment, as well as its survival, colonization, growth and development on some substrates through chemotactic movement and quorum sensing (QS) (Nan et al. 2011). *A. ferrooxidans* can tolerate organic compounds and metal ions within a certain concentration, which is one of the main reasons for its survival in extreme environments such as metal mines, coal mines, and sludge in sewage treatment plant. *A. ferrooxidans* grows and metabolizes in iron-rich environments and forms protein-coated nanoparticles, namely magnetosome, composed of inorganic Fe₃O₄ through the uptake of iron (Xie et al. 2005; Yan et al. 2012a). Although the number of magnetosomes in *A. ferrooxidans* is not large, they can allow cells to perform a certain degree of magnetotaxis, which is an effective means for cells to respond to high concentration

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of extracellular iron that could result in toxicity and to starvation that iron is deficient (Yan et al. 2013). Compared with the present difficulty in large-scale culturing of pure magnetotactic bacterial strains, large-scale culturing of *A. ferrooxidans* to obtain magnetosomes is simple.

Furthermore, *A. ferrooxidans* could effectively remove sulfur in coal, waste rubber and gas based on its oxidizing capacity of Fe^{2+} or reduced sulfur compounds. The recovery of metal from solid waste, such as low-grade ore, electronic waste, and sludge, has become the focus of intense research within metallurgy, environmental protection, and agriculture (Valdés et al. 2008). Traditional chemical, pyrometallurgical, hydrometallurgical extraction of metals from solid waste causes secondary pollution, for instance, the acid produced by these processes could cause soil and water acidification and the toxic gases generated by their combustion can pollute the atmosphere (Arshadi and Mousavi 2015). *A. ferrooxidans*-mediated leaching of metal from metalliferous solid waste is attracting the attention and interest of many investigators because of its low cost, low energy consumption and being environmentally friendly. *A. ferrooxidans* are also being used to biosynthesize biochemical production and to biomachine metal piece (Mukherjee et al. 2016; Xenophon et al. 2015).

Here, we elaborate the general biological features of *A. ferrooxidans* including morphology, genetic diversity, and genomics. The electron transfer pathways of Fe^{2+} and sulfur oxidation, and the mechanisms involved in the tolerance to small molecule organic matter and metal ions were described. We also summarized the magnetic properties of *A. ferrooxidans* and the biocompatibility of its magnetosomes. Then we present the potential application of *A. ferrooxidans* in biological desulfurization, metal recycling from metal-bearing wastes, synthesis of biochemical production, and machining of workpiece. Finally, some perspectives and possible directions for the genomics, iron/sulfur metabolic pathway, chemotaxis, and magnetosome formation of *A. ferrooxidans* are brief proposed.

General biological features

Ecological distribution and morphology

A. ferrooxidans widely distributes in natural environments, such as soil, sea water, fresh water, sediment, sulfur springs, volcanic ash, and acid mine drainage generated by various metal sulfide ores (Hedrich et al. 2011; Schrenk et al. 1998). This organism forms oval colonies of brown or orange color in 9K solid medium using Fe^{2+} as an energy source, and it forms round colonies with a yellow center and a white periphery in solid medium using $\text{Na}_2\text{S}_2\text{O}_3$ as the energy source. *A. ferrooxidans* usually

occur as single or double cells or as a chain. Its cellular morphology is related to the energy source, which results in club or rod shapes in the medium with Fe^{2+} or S as the substrate, respectively (Yan et al. 2012b). The surface of *A. ferrooxidans* contains a variety of chemical groups including CH, CH_2 , CH_3 , NH, NH_2 , NH_3 , COOH and CONH (Yan et al. 2010a).

Physico-biochemical traits

A. ferrooxidans normally grows in the pH range 1.0–4.5 with an optimum 2.0, but unlikely survives at pH above 6.5 or below 1.0 (Jensen and Webb 1995a; Valdés et al. 2008). It thrives over a wide range of temperatures (10–47 °C) in nature where ferrous ion and/or reduced sulfur are/is relatively abundant and the sodium chloride titer lowers than 10 g/L (Hedrich et al. 2011; Robertson and Kuenen 2006; Zammit et al. 2012). The difference between optimum pH values for different strains is related to the specific structure and permeability of the cell membrane surface (Schrader and Holmes 1988). *A. ferrooxidans* can utilize a variety of electron donors including Fe^{2+} , S^0 , H_2 formic acid, and reduced inorganic sulfur compounds, and either O_2 , S^0 , or Fe^{3+} as electron acceptor (Osorio et al. 2013). *A. ferrooxidans* is also capable of using electrical current as the sole energy source for respiration (Ishii et al. 2015). In addition, *A. ferrooxidans* can obtain energy for growth through oxidizing molybdenum and tetravalent uranium (DiSpirito and Tuovinen 1982; Sugio et al. 1992a).

A. ferrooxidans fixes CO_2 through the Calvin cycle and uses it as only carbon source (Valdés et al. 2008). The most suitable CO_2 concentration for bacterial growth is 7–8% (Barron and Lueking 1990). A number of enzymes associated with the Calvin cycle were found in *A. ferrooxidans*, including D-ribulose-1,5-bisphosphate carboxylase/oxygenase (RuBisCO) and phosphoribulokinase (PRK) (Esparza et al. 2010; Gale and Beck 1967). *A. ferrooxidans* can obtain nitrogen for growth through ammonia assimilation and nitrogen fixation. A hypothetical nitrogenase gene cluster (*nifH-D-K-fer1-fer2-E-N-X*) was found in *A. ferrooxidans*, which encodes the nitrogenase and proteins that are involved in nitrogenase MoCo cofactor synthesis (Valdés et al. 2003). Ammonia assimilation-related genes are also present in the genome of *A. ferrooxidans*, including the ammonia uptake-related genes *amt1*, *amt2*, and *amtB*; the nitrogen metabolism regulation-related genes *glnK-1*; and the *glnA* gene related to ammonia integration into glutamate (Valdés et al. 2008). The intake and utilization of ammonia is complete with the help of the proteins encoded by the above-mentioned genes (Dixon and Kahn 2004).

Genetic diversity

Many *A. ferrooxidans* strains have the similar phenotypic characteristics, while there exist significant differences in their total DNA composition and 16S rRNA gene sequences. Therefore, *A. ferrooxidans* can be divided into different genomovars, and the increases in the number of distinguishing phenotypic traits of the genomovars lead to the occurrence of the phylogenetically heterogeneous species and the separate strains (Karavaiko et al. 2003; Ni et al. 2007). Twenty-five *A. ferrooxidans* strains isolated from ores and ore concentrates collected in different regions of the world showed different DNA restriction profiles obtained by pulsed-field gel electrophoresis (PFGE). Based on the degree of DNA relatedness, 17 strains were divided into four genomovars, while other strains could not be assigned to any of the genomovars (Karavaiko et al. 2003).

Twelve *A. ferrooxidans* strains isolated from different geographical locations were divided into five homologous groups based on the randomly amplified polymorphic DNA (RAPD) procedure, and similarities between the strains ranged from 5.49 to 85.14% (Waltenbury et al. 2005). Twenty *A. ferrooxidans* strains derived from different environments in China and two American Type Culture Collection (ATCC) standard strains were assigned to three systematic groups (Ni et al. 2007). The lengths of the 16S–23S intergenic spacer regions (ISR) in strains derived from different groups were significantly different, and the sizes were found to be 440–441 bp (Group 1), 448–456 bp (Group 2), 452–456 bp (Group 3), respectively. The genetic diversity of *A. ferrooxidans* was not only aroused by the diversity and discrepancy of environment, but also caused by the use of different culture treatments under laboratory conditions (Kondratyeva et al. 1995).

Genome and proteome

More than 500 isolates of *A. ferrooxidans* sensu stricto have been obtained, while only eight *A. ferrooxidans* strains have been sequenced by whole-genome analysis so far (Ulloa et al. 2017). The strain ATCC23270 was sequenced and published by TIGR in 2006 and the complete sequence of strain ATCC53993 was published in 2008 (Valdés et al. 2008). A 160 kb exclusive genomic island (GI) containing several genes coding for an additional putative copper-ATPase and a Cus system is present in ATCC53993, but is absent in ATCC23270 (Martínez-Bussenius et al. 2016). The remaining six *A. ferrooxidans* genomes are incomplete sequences. The genome of *A. ferrooxidans* is a single circular chromosome of 2.89–4.18 Mb with a G+C content of 57.6–58.9%, and contains a number of pseudogenes (Table 1).

Proteomics gives insight into the regulation and function of proteins via comparing the difference of protein expression and analyzing the variety of protein properties. The different energy substrates, i.e., S and Fe²⁺, can result in different protein expression profiles for *A. ferrooxidans*. It has been reported that the expression of proteins were upregulated when Fe²⁺ is used as the energy substrate, and more than 7 proteins including rusticyanin, cytochrome c552, PstS, CbbQ protein, the small and large subunits of ribulose biphosphate carboxylase were isolated and identified (Ramírez et al. 2004). However, the expression of these proteins is down-regulated when S is used as the energy substrate (Ramírez et al. 2004). A total of 131 proteins involved in transport, conjugation, envelope, energy metabolism and conformation were isolated and identified from the periplasm of *A. ferrooxidans* cultured with Na₂S₂O₃ as the energy substrate (Chi et al. 2007). Studies show that the expression of the outer membrane protein Omp40 in *A. ferrooxidans* change with the variations of pH and phosphate concentration (Guiliani and Jerez 2000). Additionally, it has been documented that the proteome of *A. ferrooxidans*

Table 1 Comparison of genomic characteristics of several *A. ferrooxidans* strains

	Strains							
	ATCC23270	ATCC 53993	DLC-5	YQH-1	Hel18	BY0502	RV1	GGI-221
Accession	NC_011761	NC_011206	JNNH01	LJBT01	LQRJ01	LVXZ01	– ^a	AEFB01
Size (Mb)	2.98	2.89	4.18	3.11	3.11	2.98	3.07	3.17
GC %	58.8	58.9	57.6	58.6	56.8	56.8	58.5	58.6
Protein	2894	2755	4131	2916	–	2816	–	4007
Gene	3030	2862	4299	3089	3211	3141	3124	4.79
rRNA	6	6	9	10	–	52	1	–
tRNA	79	46	55	42	47	45	47	75
Pseudogene	48	53	0	64	–	273	–	–

^aNot provided

changed in response to metals such as copper, cadmium, nickel, zinc, uranium (Martínez-Bussenius et al. 2016).

Ferrous iron- and sulfur-oxidizing system

Ferrous iron-oxidizing system

The electron transfer pathway of *A. ferrooxidans* during Fe^{2+} oxidation is of particular interest to many scholars. It has been documented that electron transfer during Fe^{2+} oxidation is carried out bidirectionally in both a “downhill potential gradient” and an “uphill potential gradient”, and that the two electron transfer pathways are interconnected.

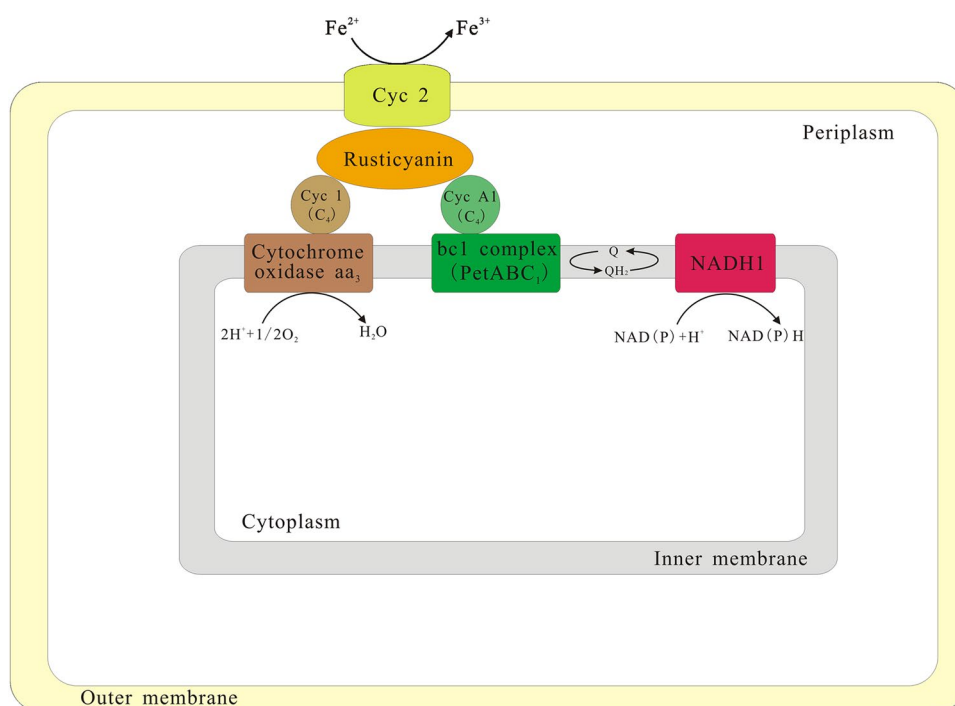
Downhill electron pathway

During the Fe^{2+} oxidation process, approximately 95% of the electrons enter the downhill electron pathway, electrons move across the cell membrane into the cytoplasm through a series of electron transport carriers, and finally transport to oxygen, and then react with protons to generate water; at the same time the protons are pumped out of the membrane, thus forming a proton gradient (Brasseur et al. 2004; Valdés et al. 2008). ATP synthase utilizes this proton gradient to form a proton motive force to synthesize ATP. The model of the downhill electron pathway can be expressed as following: $\text{Fe}^{2+} \rightarrow \text{Cyc 2} \rightarrow \text{Rus} \rightarrow \text{Cyc 1} \rightarrow \text{Cyt aa}_3 \rightarrow \text{O}_2$ (Fig. 1) (Quatrini et al. 2009; Rawlings 2005).

The coding genes involved in the downhill electron pathway of *A. ferrooxidans* are present in the single-copy *rus* operon. The operon consists of eight genes (*cyc2*, *cyc1*, *coxB*, *coxA*, *coxC*, *coxD*, *rus*) with known function and a *cup* gene with unknown function (Yarzabal et al. 2004). The expression and regulation of the *rus* operon is related to the substrate electron donor (Yarzabal et al. 2004). The operon could be expressed in a large number in cells that use Fe^{2+} as an electron donor, but only transiently expressed during early logarithmic growth of cells using S as an electron donor (Quatrini et al. 2005; Yarzabal et al. 2003).

The Cyc 2 is considered to be the first electron carrier in the electron transport chain on the outer membrane of *A. ferrooxidans* (Blake and Shute 1994). The Rus belongs to type I rusticyanin, which is located in the periplasmic space and plays a ‘bypass’ role during electron transfer (Fig. 1). The Cyc 1 is a c_4 -type cytochrome encoded by the *cyc1* gene (Yarzabal et al. 2002). The Rus and Cyc 1 can interact to form rusticyanin–cytochrome c_4 complex that allows electrons to be efficiently transferred from Rus to Cyc 1 (Campodonico et al. 2016). Cyt aa_3 is located in the intracellular membrane of *A. ferrooxidans* and is supposed to be the important terminal oxidase in electron transfer chain. It is responsible for transferring electrons from Cyc 1 to O_2 , thereby catalyzing the reduction of O_2 to H_2O and coupling with the proton pump to drive ATP synthesis.

Fig. 1 Model of Fe^{2+} oxidation electron transport pathway of *A. ferrooxidans*



Uphill electron pathway

During Fe^{2+} oxidation, 5% of the electrons enter the uphill electron pathway which can be described as follows: $\text{Fe}^{2+} \rightarrow \text{Cyc 2} \rightarrow \text{Rus} \rightarrow \text{Cyc A1} \rightarrow bc_1 \rightarrow \text{QH}_2/\text{Q} \rightarrow \text{NDH-1} \rightarrow \text{NAD}^+$ (Fig. 1). Here, Cyc 2, Rus, Cyc A1, bc_1 , QH_2 , Q and NDH-1 refer to cytochrome *c*, rusticyanin, cytochrome c_4 , cytochrome bc_1 , quinol, quinone and NADH dehydrogenase-1, respectively (Yamanaka 2008).

There are two bc_1 complexes in *A. ferrooxidans*. The bc_1 complex encoded by *petI* is responsible for the inverse electron transfer when Fe^{2+} is available as a substrate, whereas the bc_1 complex encoded by *petII* is the one functioning in forward electron transfer when S is available as a substrate (Brasseur et al. 2002). The *resB* and *resC* genes in the *res* operon, which is adjacent to the *pet* operon, encode ResB and ResC protein, respectively. These proteins may play a role as a molecular chaperone in the maturation of the c_1 cytochrome of the bc_1 complex (Quatrini et al. 2009).

Sulfur-oxidizing system

A. ferrooxidans can obtain the energy required for growth by oxidizing reduced sulfur such as S^0 , S^{2-} , $\text{S}_2\text{O}_3^{2-}$, SO_3^{2-} and $\text{S}_4\text{O}_6^{2-}$. Several enzymes including thiol-bearing outer-membrane protein, sulfur dioxygenase, sulfite-oxidoreductase, and sulfide-quinone oxidoreductase, thiosulfate oxidase, rhodanase, and tetrathionate hydrolase are involved in oxidation of reduced inorganic sulfur (Holmes and Bonnefoy 2007). However, only a minor portion of them are coupled to the respiratory chain, and these couplings occur at different nodes within the chain (Quatrini et al. 2006). The electrons obtained from soluble sulfur (H_2S , $\text{S}_2\text{O}_3^{2-}$), or sulfane sulfate (GSSH) which converted from insoluble sulfur enter the respiratory chain at the level of the quinone pool. These electrons can then either be transferred (1) directly to terminal oxidases *bd* or *bo*₃, or indirectly to the *aa*₃ oxidase, or (2) to NADH complex I to produce a reducing power (Fig. 2) (Quatrini et al. 2009).

Two mechanisms for elemental sulfur oxidation by *A. ferrooxidans* have been proposed: an aerobic oxidation in which the sulfur is oxidized to sulfite using oxygen as terminal

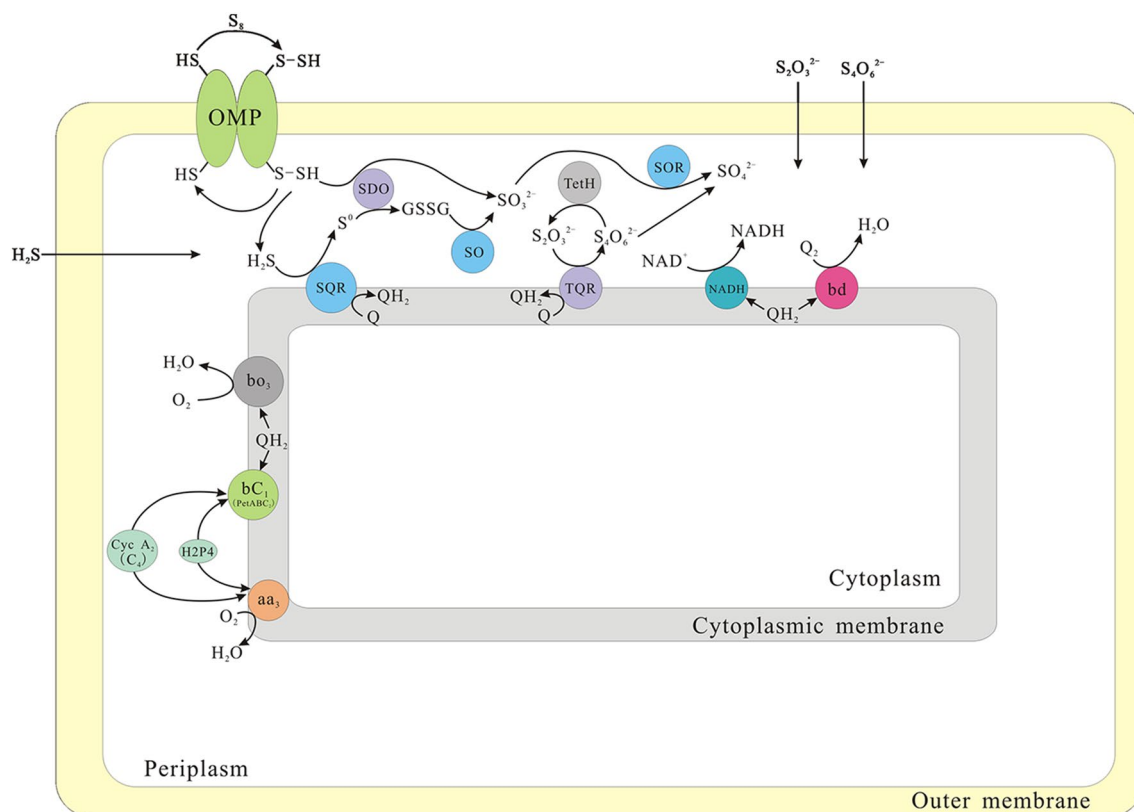


Fig. 2 Model of sulfur oxidation electron transport pathway of *A. ferrooxidans*. SQR sulfide quinone reductase, TQR thiosulfate quinol reductase, OMP thiol-bearing outer membrane protein, SDO sulfur

dioxygenase, SO sulfur oxygenase, SOR sulfur oxygenase reductase, TetH tetrathionate hydrolase, HIPI high potential iron–sulfur protein

electron acceptor, and an anaerobic oxidation in which the hydrogen sulfide-ferrous ion oxidoreductase oxidizes elemental sulfur with Fe^{3+} as terminal electron acceptor to produce sulfite (Holmes and Bonnefoy 2007; Valdés et al. 2008). There are three pathways to oxidize sulfite into sulfate for *A. ferrooxidans*. In pathway A, the sulfite-cytochrome *c* oxidoreductase containing molybdenum directly oxidizes sulfite to sulfate and the released electrons enter the respiratory chain through coenzyme and couple to oxidative phosphorylation (Sugio et al. 1992b). Pathway B is the AMP-dependent oxidation via the intermediate adenylylsulfate (APS). In this pathway, the APS reductase catalyzes the AMP and sulfate to form APS, after which the AMP moiety of APS is transferred either to pyrophosphate (PPi) to form ATP by ATP sulfurylase or to phosphate (Pi) to generate ADP via APS-phosphate adenylyltransferase (APAT), and the sulfate release from APS at the same time (Kappler and Dahl 2001). In pathway C, sulfite-ferrous ion oxidoreductase catalyzes the oxidation of sulfite with Fe^{3+} as an electron acceptor to generate sulfate and Fe^{2+} (Sugio et al. 1992b).

Chemotactic movement and quorum sensing

A. ferrooxidans facilitates the use of beneficial substances through positive chemotaxis, and avoids harmful substances through negative chemotaxis. In addition, *A. ferrooxidans* monitors its cell-population density in the environment using quorum sensing. Some particular genes of *A. ferrooxidans* are repressed at low cell density, leading to an uncommunicative or do-noting state for bacteria. When the number of cells increases, the quorum sensing system changes and coordinates behavior among the bacterial cells of *A. ferrooxidans*, resulting in several special physiological functions that a single cell cannot engage in. The chemotaxis and quorum

sensing play important roles in bacteria–mineral interactions during bioleaching.

Chemotactic movement

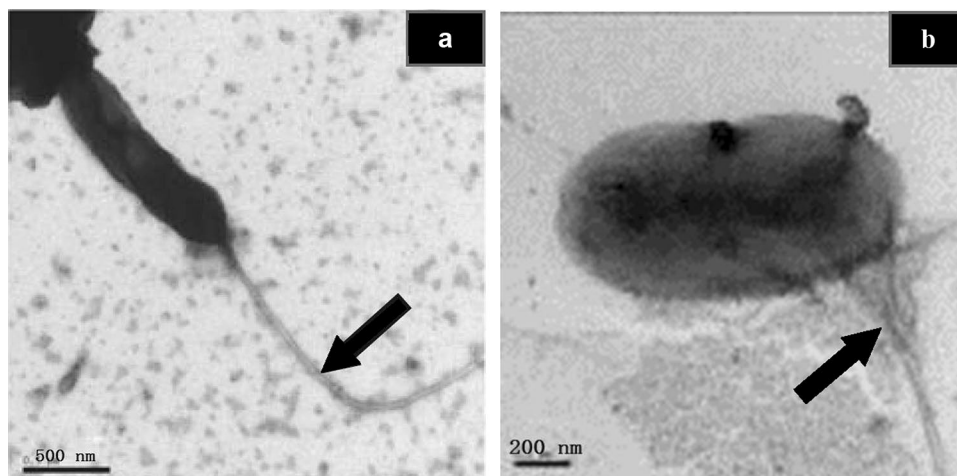
A. ferrooxidans shows positive chemotaxis towards $\text{Na}_2\text{S}_2\text{O}_3$, chalcopyrite, and pyrite (Chakraborty and Roy 1992; Jerez 2001). A variety of chemotaxis-related *che* genes widely distributed within the genome of *A. ferrooxidans*. It has been reported that the genes encoding the chemotactic protein CheY were found in different *A. ferrooxidans* strains. *A. ferrooxidans* swims in liquid by rotating its flagella with a diameter of about 16 nm (Fig. 3a) (Li et al. 2010b). A 40-kDa flagellin contained thiol groups was found to mediate the strong adhesion between *A. ferrooxidans* cells and sulfur (Ohmura et al. 1996). Contradictorily, the *fla* or *fla*-related genes encoding flagella were not identified in the complete genomic sequence of *A. ferrooxidans*. The reason of this discrepancy may be that the *fla* genes have been lost in special culture utilized for sequencing (Valdés et al. 2008).

Several strains of *A. ferrooxidans* can produce type IV pili (Fig. 3b) which are required for bacterial sliding, twitching motility, and adherence (Li et al. 2010b). *A. ferrooxidans* contains four genes, *pilV*, *pilW*, *pilU*, and *pilT*, which encode type IV pilin. It has been documented that the type IV pili of *A. ferrooxidans* has detectable conductivity and serve as biological nanowires mediating electrons transfer from Fe^{2+} oxides surface to the cell surface (Li and Li 2014).

Quorum sensing

Quorum sensing (QS) is a bacterial communication system which enables bacteria to synchronize their behavior on a population-wide scale (Bassler 2016). The functional type

Fig. 3 Transmission electron micrographs of flagella (a) and pili (b) from *A. ferrooxidans*. Figure a and b reproduced from Li et al. (2010b) with permission from Springer



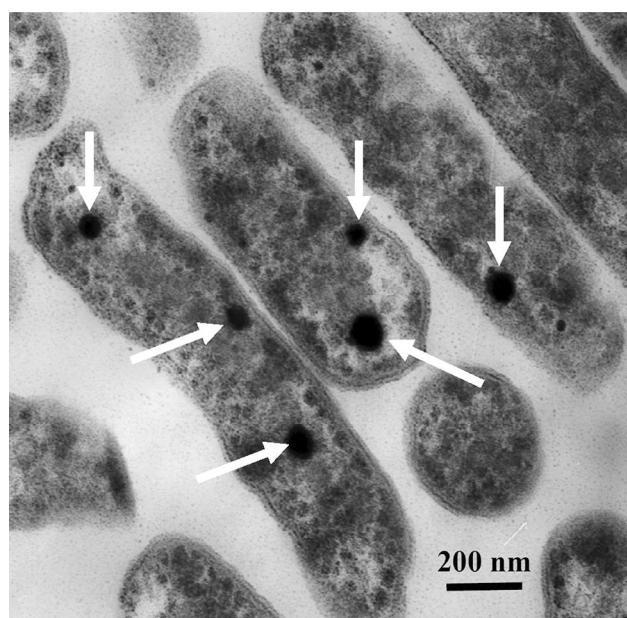


Fig. 4 Transmission electron micrographs of magnetosome from *A. ferrooxidans*. The figure reproduced is from Yan et al. (2012a) with permission from Elsevier

AI-1 QS system of *A. ferrooxidans* is similar to that of other gram-negative bacteria and consists of four molecular elements: an AHL synthase (AfeI); a transcriptional regulator (AfeR), which is capable of binding to AHL; an *afe*-box targeted for the binary complex (AfeR-AHL); and the various AHL signaling molecule generated by AfeI (González et al. 2013). AfeI, as the main producer of AHL, is able to synthesize nine different AHL molecules with diverse C3 substituent groups (hydroxyl or oxo) (Farah et al. 2005). A synthetic superagonist has been proved to enhance the transcription level of *afeI* gene in sessile *A. ferrooxidans* cells, which not only led to the stabilization of the biofilm but also to the synthesis of various AHL molecules (Mamani et al. 2016). The type AI-1 QS system of *A. ferrooxidans* plays an important role in bioleaching, which involves the biosynthesis of extracellular polymers and bacterial adhesion to minerals (Banderas and Guiliani 2013).

Additionally, a second AHL synthetase, Act, has been identified in *A. ferrooxidans*. Its encoding gene, *act*, expressed higher expressed at higher levels when *A. ferrooxidans* cultured in Fe^{2+} substrate medium than when it grown in sulfur substrate medium (Rivas et al. 2007). A Lux-like QS system was also be found in *A. ferrooxidans* (Rivas et al. 2005). It has been reported that this system might control the biofilm formation (Farah et al. 2005). Therefore, *A. ferrooxidans* contains two QS systems, one based on Act and the other based on Lux-like proteins. The function of these two QS systems may rely on environmental signals related

to the ability of *A. ferrooxidans* to colonize and utilize iron- and sulfur-containing ores (Rivas et al. 2007).

Magnetic property

The membrane-bound magnetic iron-bearing inorganic nanocrystal magnetites, named magnetosomes, are synthesized in *A. ferrooxidans* through uptake iron from niche (Fig. 4) (Yan et al. 2012a). The magnetosome with diameters between 15 and 70 nm is the origin of the weak magnetotaxis of *A. ferrooxidans*, and might serve as iron storage and detoxification compartments (Biologie 2011; Yan et al. 2013). *A. ferrooxidans* shows magnetotaxis on the semisolid plates and in liquid medium under microscope. Fe^{2+} is not only the energy source for *A. ferrooxidans*, but also the iron source for synthesis of magnetosomes in cells (Zhang et al. 2012). The concentration of Fe^{2+} influences the expression of the genes related to magnetosome production including *thy*, *magA*, *mpsA* and *mamB*, thus affecting the formation of magnetosomes (Liu et al. 2008). *A. ferrooxidans* contains a large amount of iron, reaching up to 1.24% of the cell dry weight, which may be the potential raw material for magnetosome synthesis (Zhang et al. 2012). Additionally, oxygen concentration, nitrogen source titer, and magnetic field intensity were also found to affect the formation of magnetosomes of *A. ferrooxidans*. The iron content and the average number of magnetosomes in *A. ferrooxidans* can be elevated via optimization, which increase from 11.87 to 25.34 mg/g, and from 2 to 5 per cell, respectively (Yan et al. 2016).

The pot-bellied shape hysteresis loop has been observed at room temperature in the calcined cell sample of *A. ferrooxidans* (Fig. 5) (Yan et al. 2013). The coercivity (B_c) of 16.36 mT which is within the range of 7.5–46.7 mT suggests

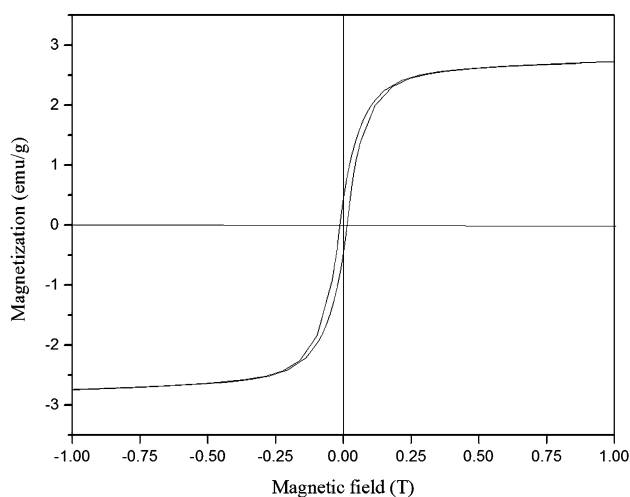


Fig. 5 Hysteresis loop of calcined cells of *A. ferrooxidans*. The figure is reproduced from Yan et al. (2013) with permission from Elsevier

a similar room temperature magnetic feature between *A. ferrooxidans* and other magnetotactic bacteria (Pósfai et al. 2006). Saturation magnetization (M_s) values for pure magnetite and *A. ferrooxidans* are 92.40 and 2.64 emu/g, respectively (Weiss et al. 2004; Yan et al. 2013). Therefore, the magnetite content of *A. ferrooxidans* can be calculated directly and its value is 2.86%, which is significantly higher than that obtained in other magnetotactic bacteria (Pósfai et al. 2006; Yan et al. 2013). The Curie temperature of 454 °C for the calcined cells of *A. ferrooxidans* is found to be significantly lower than that of pure magnetite (580 °C) due to the presence of impurities in calcined cells (Nowaczek et al. 2000; Yan et al. 2013). The Verwey transition temperature (T_v) obtained from the bifurcation of the field cooled (FC) and zero field cooled (ZFC) warming curves is about 100 K, indicating the presence of magnetite in the sample of *A. ferrooxidans* (Popa et al. 2009; Yan et al. 2013). The δ_{FC}/δ_{ZFC} ratio, an indicator of the arrangement of magnetosomes in bacteria, is found to be 1.27 for *A. ferrooxidans*, which suggests that the magnetosomes distribute dispersely in bacteria (Li et al. 2010a; Yan et al. 2013).

There are some carboxyl, hydroxyl and amino groups on the surface of the magnetosome extracted from *A. ferrooxidans* (Yan et al. 2012a). The nanoscale magnetite particles inside magnetosomes of *A. ferrooxidans* have high crystallographic perfection (Yan et al. 2013). It has been documented that the magnetosomes are not able to cytotoxicity, genotoxicity and hemolysis up to the concentration of 4.0 mg/mL (Yan et al. 2012a). Based on this good biocompatibility of magnetosome and the non-fastidious culture condition for bacteria, *A. ferrooxidans* should be an excellent candidate for the mass production of magnetosomes.

Potential application of *A. ferrooxidans*

Bioremoval of sulfur from solids and gases

A. ferrooxidans can remove inorganic sulfur in the form of FeS_2 in coal by contact and non-contact pathways (Dastidar et al. 2000; Rai 1985). In the contact pathway, *A. ferrooxidans* directly transforms FeS_2 to Fe^{3+} and SO_4^{2-} . In the non-contact pathway, FeS_2 is converted by Fe^{3+} to form SO_4^{2-} or S, and the generated Fe^{2+} is immediately oxidized to Fe^{3+} by *A. ferrooxidans* (Hong et al. 2013; Rai 1985). The removal technologies for sulfur from coal using *A. ferrooxidans*, including batch, fed-batch, fluidized-bed, packed-bed, airlift reactor and heap leaching, can be found anywhere in the literature (Hong et al. 2013; Mannivannan et al. 2008). *A. ferrooxidans* has a good desulfurization effect on the waste rubber powder. The S–S bond in pulverized waste rubber can be gradually oxidized by *A. ferrooxidans* to form a sulfoxide or sulfo functional groups, and finally separates

from the rubber substrate in the form of SO_4^{2-} (Jiang et al. 2010). It has been reported that the sulfur content in waste rubber decrease by more than 13.9% after treatment using *A. ferrooxidans* (Wang et al. 2011).

The SO_2 or H_2S in gas can be transformed to form S or SO_4^{2-} by *A. ferrooxidans* or Fe^{3+} generated from bacterial oxidization of Fe^{2+} (Jensen and Webb 1995b). The fixed bed and biofilter inoculated with *A. ferrooxidans* can effectively remove SO_2 with desulfurization rates of up to 96.0 and 99.0%, respectively (Huang 2009). It has been reported that the single bioreactor using free *A. ferrooxidans* can remove 98.0% SO_2 or 99.0% H_2S from gas, respectively (Yu et al. 2007). The continuous bio-chemical system consisted of immobilized *A. ferrooxidans* bioreactor and chemical reactor has been found to effectively remove 99.0% SO_2 or 95.0% H_2S from gas, respectively. *Acidithiobacillus thiooxidans* (*A. thiooxidans*), one specie of the genus *Acidithiobacillus*, is considered to play an important role in desulfurization (Lee et al. 2006). *A. thiooxidans*-driven desulfurization methods including batch, fed-batch, packed-bed and stirred-tank reactor were used to remove H_2S from gas (Aroca et al. 2007; Lee et al. 2006; Oprime et al. 2001). The desulfurization efficiency were found to be in the range of 91.0–99.0%, which were of the same order of magnitude or higher than that reported for *A. ferrooxidans* (Aroca et al. 2007; Lee et al. 2006; Oprime et al. 2001; Yu et al. 2007). These indicate that both *A. ferrooxidans* and *A. thiooxidans* have considerable potential for the bioremoval of from H_2S from gas.

Bioleaching of metals from metal-bearing wastes

Bioleaching is defined as the solubilization of metal cations from insoluble solids by sulfur- and/or iron-oxidizing bacteria or by the products of bacterial metabolism (Yan et al. 2017a). *A. ferrooxidans* is considered as an important member of bioleaching bacteria used to recover metal. It has been used in different biomining process including in-suit bioleaching, heap/dump bioleaching and tank bioleaching (Brierley 2008). The possible mechanisms including contact, non-contact and cooperative mechanisms have been postulated to explain the bioleaching of metals from sulfide ores by *A. ferrooxidans* (Mahmoud et al. 2017; Rohwerder et al. 2003). In the course of the contact mechanism, the *A. ferrooxidans* cells attach to the surface of sulfide minerals mediated by Fe^{3+} ions, the resulting complexed Fe^{3+} ions begin to decompose the metal sulfide (Fig. 6a) (Mahmoud et al. 2017; Sampson et al. 2000). The non-contact mechanism assumes that the free Fe^{3+} ions oxidized by planktonic *A. ferrooxidans* cells attack metal sulfides and are reduced to Fe^{2+} ions (Fig. 6b) (Mahmoud et al. 2017). The cooperative leaching refers to the dissolving of metal sulfide by

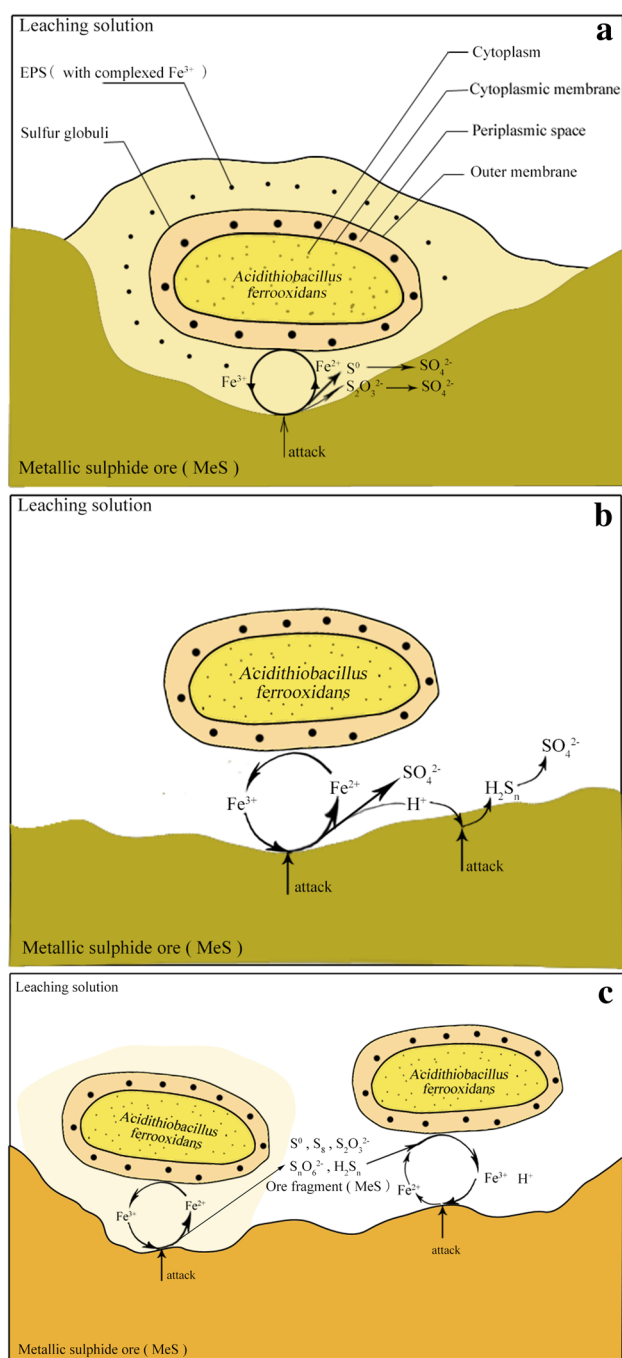


Fig. 6 Models for the proposed contact (a), non-contact (b), and cooperative (c) mechanism catalyzed by *A. ferrooxidans*

attached *A. ferrooxidans* cells on the mineral surface and by planktonic bacteria, which in essence is the combination of contact and non-contact leaching (Fig. 6c).

A. ferrooxidans has a potential for application in leaching metals including copper, zinc, nickel, uranium, arsenic, and cobalt from low-grade ores (Table 2) (Dave et al. 2008; Dong et al. 2013; Haghshenas et al. 2012; Mousavi et al. 2006; Nguyen et al. 2015; Pal et al. 2010; Patel et al. 2011; Pogaku

and Kodali 2006; Yan et al. 2017a). The metal leaching rate is found to be related to the affinity between *A. ferrooxidans* cells and minerals. *A. ferrooxidans* can extract 95, 85, 54, 67 and 18% of copper from djurleite, bornite, covellite, massive sulfide copper and porphyry chalcocopyrite, respectively (Dong et al. 2013). Zinc dissolution from Iranian low-grade complex zinc–lead ore reached 51.0% with *A. ferrooxidans* (Mousavi et al. 2006). The bioleaching of zinc from zinc-containing ores by *A. ferrooxidans* can be efficiently performed at a temperature range of 26–35 °C, a pH range of 1.2–3.0 and an initial Fe^{2+} titer range of 2.0–12.0 g/L (Gupta et al. 2003; Haghshenas et al. 2009a, b). Zinc also can be extracted by *A. ferrooxidans* from sphalerite which is considered as one of the most important sources of zinc production (Haghshenas et al. 2012; Xia et al. 2009). The recovery of copper and zinc by *A. ferrooxidans* from polymetallic bulk concentrate have been successfully achieved in batch processing system (Tipre and Dave 2004; Tipre et al. 2004). It has been reported that tryptic soy- and peptone-resistant thermotolerant *A. ferrooxidans* SRDSM2 can leach more than 85% copper and zinc from polymetallic concentrate in the presence of 0.5 g/L TSB (tryptone soya broth) and 2% ferrous sulfate (Patel et al. 2011).

Nickel is efficiently leached from heazlewoodite by *A. ferrooxidans* in the presence of Fe^{2+} , and a 35.0% leaching rate is achieved (Giaveno and Donati 2001). The adapted *A. ferrooxidans* can recover more than 40.0% of nickel from pentlandite and other nickel sulfides by using heap or tank bioleaching system (Watling 2008). *A. ferrooxidans* can leach 98.0 and 96.8% of uranium from Spanish uranium ore and granite porphyry, respectively (Muñoz et al. 1995; Qiu et al. 2011). *A. ferrooxidans* is able to modify realgar surfaces and to efficiently enhance the dissolution of arsenic (Chen et al. 2011; Yan et al. 2017a). It has been reported that 82.5% of arsenic can be recovered from complex galena concentrate in fluidized bed system inoculated with *A. ferrooxidans* (Makita et al. 2004). Additionally, *A. ferrooxidans* can leach cobalt from a low grade nickel–copper–cobalt sulfide ore, the leaching rate of cobalt achieve to 62% after an 18 d acid pretreatment and 50 d microbial leaching treatment (Yang et al. 2011).

The potential application of *A. ferrooxidans* towards recycling metals from printed circuit boards (PCB), cell/battery and waste electric cables has already been documented (Table 2) (Bajestani et al. 2014; Jadhav and Hong 2013; Lambert et al. 2015; Liang et al. 2016; Yang et al. 2017). Bioleaching of metals from PCB primarily focuses on copper leaching due to the high content of copper in PCB. *A. ferrooxidans* can recover more than 94.8% of copper from PCB in batch or tank leaching system (Chen et al. 2015; Yang et al. 2014). The leaching efficiency of copper from PCB was markedly enhanced but with shortened leaching periods by dissolved oxygen-shifted strategy (Liang et al.

Table 2 Examples of metal-bearing wastes treated with *A. ferrooxidans*

Wastes	Leaching technique and condition	Strain	Leaching rate	References
Djurleite	Batch (flask): pH 2.0, <i>T</i> 30 °C, Fe(II) 1.5 g/L, pulp density 2.0% (w/v), shake speed 160 rpm	LD-1	Cu 95.0%	Dong et al. (2013)
Bornite	Batch (flask): pH 2.0, <i>T</i> 30 °C, Fe(II) 0 g/L, pulp density 2.0% (w/v), shake speed 160 rpm	LD-1	Cu 85.0%	Dong et al. (2013)
Complex zinc–lead ore	Batch (flask): pH 1.5, <i>T</i> 33 °C, Fe(II) 7.0 g/L, pulp density 2.0% (w/v), shake speed 180 rpm	– ^a	Zn 51.0%	Mousavi et al. (2006)
Sphalerite	Batch (flask): pH 1.94, <i>T</i> 33.7 °C, Fe(II) 7.4 g/L, pulp density 4.0% (w/v), shake speed 210 rpm	PTCC 1647	Zn 85.0%	Haghshenas et al. (2012)
Mine tailings	Batch (flask): pH 2.5, <i>T</i> 30 °C, Fe(II) 9.0 g/L, pulp density 5.0% (w/v), shake speed 200 rpm	–	Mn 99.5%, Zn 78.4%, As > 60%, Cu 31.5%	Nguyen et al. (2015)
Polymetallic bulk concentrate	Batch (flask): pH 1.8, <i>T</i> 32 °C, Fe(II) 7.3 g/L, pulp density 15.0% (w/v), shake speed 150 rpm	SRDSM2	Zn 96.7%, Cu 72.2%	Patel et al. (2011)
Copper flotation concentrate	Batch (flask): pH 2.3, <i>T</i> 30 °C, Fe(II) 7.3 g/L, pulp density 10.0% (w/v), shake speed 140 rpm	Tf-44	Ni 55.0%, Co 64.0%	Pogaku and Kodali (2006)
Low-grade uranium ore	Batch (flask): pH 2.0, <i>T</i> 30 °C, Fe(II) 9.0 g/L, pulp density 10.0% (w/v), shake speed 150 rpm	–	U 49.0%	Pal et al. (2010)
Realgar	Batch (flask): pH 1.74, <i>T</i> 30 °C, Fe(II) 3.68 g/L, pulp density 0.95% (w/v), shake speed 150 rpm	BYQ-12	As 74.3%	Yan et al. (2017a, b)
Refractory gold ore	Batch (flask): pH 1.5, <i>T</i> 30 °C, Fe(II) 9.0 g/L, pulp density 2.0% (w/v), shake speed 150 rpm	HGM 5	As 39.3%	Dave et al. (2008)
PCB	Batch (flask): pH 2.25, <i>T</i> 30 °C, Fe(II) 9.0 g/L, pulp density 0.4% (w/v), shake speed 160 rpm	Z1	Cu 92.6%, Al 85.2%, Zn 95.2%	Yang et al. (2017)
PCB	Batch (fermentor): pH 1.5, <i>T</i> 30 °C, Fe(II) 2.0 g/L, pulp density 1.8% (w/v), controlling DO at 10% for first 64 h and switching to 20% afterwards	JSTU-02	Cu 94.1%	Liang et al. (2016)
Button cell battery	Batch (flask): pH 2.5, <i>T</i> 30 °C, Fe(II) 7.3 g/L, pulp density 0.1% (w/v), shake speed 150 rpm	BCRC 13820	Ag 98.0%	Jadhav and Hong (2013)
Spent household batteries	Batch (flask): pH 1.0, <i>T</i> 30 °C, Fe(II) 9.0 g/L, pulp density 1.0% (w/v), shake speed 160 rpm	PTCC 1647	Ni 87.0%, Cd 67.0%, Co 93.7%	Bajestani et al. (2014)

Table 2 (continued)

Wastes	Leaching technique and condition	Strain	Leaching rate	References
Waste electric cables	Batch (fermentor): pH 1.9, <i>T</i> 35 °C, Fe(II) 1.0 g/L, pulp density 9.0% (w/v), stirred speed 700 rpm	–	Cu > 90.0%	Lambert et al. (2015)
Sewage sludge	Batch (flask): pH 4.0, <i>T</i> 28 °C, Fe(II) 4.0 g/L, pulp density 1.8% (w/v), shake speed 150 rpm	PTCC 1646	Cd 71.9%, Mn 92.5%, Zn 89.1%	Ghavidel et al. (2018)
Dewatered metal-plating sludge	Batch (flask): pH 1.0, <i>T</i> 30 °C, Fe(II) 9.0 g/L, Fe(III) 1 g/L, pulp density 0.9% (w/v), shake speed 160 rpm	PTCC 1626	Cr 55.6%, Ni 58.2%	Rastegar et al. (2014)
Mercury-polluted soil	Batch (flask): pH 2.5, <i>T</i> 30 °C, Fe(II) 6.0 g/L, pulp density 1.0% (w/v), shake speed 100 rpm	SUG 2-2	Hg 92.0%	Takeuchi et al. (2001)
Contaminated soil	Batch (flask): pH 2.5, <i>T</i> 30 °C, Fe(II) 7.3 g/L, pulp density 5.0% (w/v), shake speed 150 rpm	–	Cd, Co, Cu, Ni > 50.0%	Bosecker (1999)
Contaminated sediment	Batch (flask): pH 4.0, <i>T</i> 30 °C, sulfur 5.0 g/L, pulp density 5.0% (w/v), shake speed 170 rpm	DSMZ 11477	Zn 87%, Cu 44%, Cr 12.0%	Akinci and Guven (2011)
Oil-fired ash	Batch (flask): pH 1.0, <i>T</i> 32 °C, sulfur 1.0 g/L, pulp density 5.0% (w/v), shake speed 30 rpm	PTCC 1626	Cu 87%, Ni 86.0%, V 82.0%	Rastegar et al. (2015)

^aNot provided

2016). The two-step bioleaching process (pre-culturing and leaching steps) and the microbial immobilized technology demonstrably improve the leaching efficiency of metals from PCB (Nie et al. 2015; Yang et al. 2017). It was suggested that application of graphene to catalytically enhancing copper bioleaching from PCBs is being feasible (Gu et al. 2017). Approximately 98.0% silver is dissolved from spent silver oxide-zinc button cells by using *A. ferrooxidans* (Jadhav and Hong 2013). It has been reported that *A. ferrooxidans* is able to leaching heavy metals from spent household batteries, and the recovery efficiencies is found to be 87, 67, and 93.7% for Ni, Cd and Co, respectively (Bajestani et al. 2014). The cobalt dissolution from spent lithium batteries can be improved in the presence of silver or copper ions (Zeng et al. 2012, 2013). *A. ferrooxidans* also can recover more than 90.0% copper from waste electric cables (Lambert et al. 2015). Iron oxidizing consortium consisted of *A. ferrooxidans* and/or *Leptospirillum ferriphilum* and other iron-oxidizing bacteria were also used to leaching metals from PCB. It has been reported that 85.26 and 99.99% Cu, 96.75 and 99.49% Zn and 93.23 and 84.21% Ni were extracted from c-PCB and m-PCB within 6–8 days via two-step

bioleaching process, respectively (Shah et al. 2014). The optimization of operating parameter and addition of chelating agents can increase the percent metal extraction and shorten the operation during bioleaching PCB by two-step process (Shah et al. 2015).

A. ferrooxidans has been successfully applied for extracting metals from sewage sludge, contaminated soil and sediments (Table 2) (Akinci and Guven 2011; Bosecker 1999; Ghavidel et al. 2018; Rastegar et al. 2014; Takeuchi et al. 2001). It can leach 83.0% of manganese, 77.5% of zinc, 24.0% of copper, and 10.0% of aluminum from sewage sludge (Lombardi and Garcia 2002). Adaptation of *A. ferrooxidans* to the dewatered metal-plating sludge can increase tolerance of bacterial cells to the toxicity, which led to a significant enhancement in metal recovery (Rastegar et al. 2014). More than 70% of the arsenic was dissolved by *A. ferrooxidans* from the soil containing 4.6–5.6 g/kg of arsenic (Ko et al. 2013). It has been documented that *A. ferrooxidans* significantly enhance the extraction of cadmium and nickel from contaminated sediments, and the leaching efficiencies are 80.0% for cadmium and 60.0% for nickel, respectively (Kim et al. 2005). The oil-fired ash which is a

major by-product of thermal power plants contains metals and can be treated with adapted *A. ferrooxidans* (Table 2) (Rastegar et al. 2015).

Biosynthesis of biochemical production

A. ferrooxidans can drive metabolic energy from iron oxidation catabolic process. Therefore, the $\text{Fe}^{2+}/\text{Fe}^{3+}$ redox couple could potentially provide an efficient way for using renewable energy to drive the metabolism for microbiological reduction of CO_2 (Li et al. 2014). Based on this assumption, two heterologous biosynthetic pathways were introduced into *A. ferrooxidans* to produce exogenous chemicals (isobutyric acid or heptadecane) from CO_2 using Fe^{2+} oxidation at low pH (Kernan et al. 2016). This genetically modified bacteria can produce 12 ± 1 and 4.4 ± 2.7 mg/mol iron in batch cultures and in continuous chemostat cultures, respectively (Kernan et al. 2016). By operating at high pH and high Fe^{2+} concentration with the addition of iron chelator (gluconate), the yields of isobutyric acid markedly improved and were found to be 16 ± 1 and 11 ± 1 mg/mol iron in batch and chemostat systems, respectively (Li et al. 2016).

A. ferrooxidans not only oxidizes Fe^{2+} , but also plays an important role in biomineralization which involves biosynthesis of iron-oxyhydroxysulfate materials such as jarosite, schwertmannite and akaganeite. These materials have gained increasing interest in microbiological, environmental, geological, and metallurgical fields, because they were regarded as the efficient scavengers for heavy metals and some toxic ion (Zhu et al. 2013), the Fenton-like catalysts for the degradation of organic pollutants (Yan et al. 2017b), and the excellent mineral pigments used in mural paintings and civil engineering (Wang et al. 2013). Jarosite $[(\text{Na}^+, \text{Ag}^+, \text{K}^+, \text{NH}_4^+, \text{H}_3\text{O}^+)\text{Fe}_3(\text{SO}_4)_2(\text{OH})_6]$ is one of the isostructural jarosite-alunite group minerals (Hou et al. 2015). Najarosite were synthesized by immobilized *A. ferrooxidans* in the modified mineral salt medium containing 200 mM Fe^{2+} (Wang et al. 2013). Previous researchers have successfully produced Ag-jarosite from ferric sulfate solutions obtained by oxidation of 120 mM Fe^{2+} by *A. ferrooxidans* at 22 ± 2 °C followed by 40 mM AgNO_3 addition (Mukherjee et al. 2016). The ternary (K, NH_4 , H_3O)-jarosites were also synthesized from *A. ferrooxidans* cultures in simulated bioleaching solutions (Sandy Jones et al. 2014). Schwertmannite $[\text{Fe}_8\text{O}_8(\text{OH})_{8-2x}(\text{SO}_4)_x, 1 < x < 1.75]$, a mineral with high surface area and surface reactivity, has been biosynthesized through oxidation of 144 mM Fe^{2+} by *A. ferrooxidans* LX5 cell suspensions incubated at 28 °C (Yan et al. 2017b). Akaganeite (β - FeOOH) was also successfully synthesized using ferric solution produced by the oxidation of ferrous chloride by the strain LX5 (Xiong et al. 2008).

Additionally, *A. ferrooxidans* can generate a complex ferric biogenic flocculent containing biogenic secondary minerals and biopolymeric substances, which was for sludge conditioning to improve the dewaterability of sewage sludge (Kurade et al. 2016; Murugesan et al. 2014). The moisture content of sludge was found to be reduced to 70% after treatment with biogenic flocculant produced by *A. ferrooxidans* (Kurade et al. 2016).

Biomachining of metal pieces

The utilization of microbes to remove metal from a workpiece is referred to as biological machining or biomachining. The formation of microstructures on a workpiece are carried out by microorganism-mediated metal removal or dissolution. An effective and sustainable alternative for micro-component demands, biomachining can avoid distorting or damaging the workpiece, and can ensure precision, high quality, less heat and minimum stress (Xenofontos et al. 2015). *A. ferrooxidans* has been widely applied in the biomachining of metal pieces (Jadhav and Hocheng 2016; Wang et al. 2014; Xenofontos et al. 2015). *A. ferrooxidans* can act as a catalyst by oxidizing Fe^{2+} to Fe^{3+} , the Fe^{3+} is able to oxidize pure metal (M^0) to metal ion (M^{2+} or M^{3+}) and is reduced back to Fe^{2+} , then Fe^{2+} can be reoxidized by bacteria (Xenofontos et al. 2015). Therefore, a circulatory system is formed and a workpiece can be machined by the Fe^{3+} .

The copper sheets were efficiently machined by *A. ferrooxidans* under controlled condition (pH 1.8, temperature 20 °C, vortex speed 2000 rpm) and employing appropriate inoculation strategy. The maximum Cu removal rate of 10.77 mg/cm² was achieved after 9 h biomachining at regulated pH 1.8 (Xenofontos et al. 2015). Large pits with a diameter of 450–500 μm and a depth ranging from 60 to 170 μm were detected on C1010 steel surfaces after machining 7 days by *A. ferrooxidans* inoculum, and the corrosion rate can be achieved to 3.54 ± 0.091 mm/year in 24 h (Wang et al. 2014). The copper–nickel (Cu–Ni) alloy was also successfully machined by *A. ferrooxidans* with a corrosion rate of 3.69 ± 0.26 mm/year (Jadhav and Hocheng 2016).

Perspectives and directions for future research

Although significant progress has been made on *A. ferrooxidans* studies, there are a number of theoretical and practical issues that remain need to be explored and clarified. It can be hoped that the notable breakthrough will be achieved in the following aspects.

1. The genomic data of *A. ferrooxidans* are not comprehensive and complete; it is necessary to supplement more complete genomic sequences to promote the development and deepening of pan-genomics and micro-evolutionary biology.
2. The elucidation of the metabolic pathways of iron and sulfur in *A. ferrooxidans* is a long and arduous task. Despite a few genes and proteins involved in iron and sulfur oxidation have been characterized, the information of the genes and proteins coupled iron and sulfur metabolism, and their metabolic coupling metabolism are still lacking. In addition, the processes and mechanisms of carbon and nitrogen fixation by *A. ferrooxidans* require further study.
3. Systematic and concerted efforts to explore and functionally validate the chemotactic genes and proteins are urgently required. Investigations into quorum sensing of *A. ferrooxidans* can further reveal the interaction mechanism between bacteria and environment and the mechanism of intercellular information transmission, as well as provide guidance for the regulation of *A. ferrooxidans* in industrial and environmental applications.
4. The investigation of the formation and regulation mechanism of magnetosome in *A. ferrooxidans* should be performed with increased efforts. Once the mechanism of magnetosome synthesis has been illuminated, the use of genetic engineering techniques could result in much higher yields of magnetosomes.

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Compliance with ethical standards

Ethical statement This manuscript is original and has not been submitted elsewhere for publication, in whole or in part, and no data was fabricated or manipulated (including images) in this work. The submission has been received explicitly from all co-authors. All the authors listed have contributed sufficiently to the scientific work and therefore share collective responsibility and accountability for the results.

Conflict of interest We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted. Informed consent was obtained from all individual participants included in the study.

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